POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Spring

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a). Differentiate between Von-Neumann and Harvard Architecture. Explain fetch, decode and execute cycle in microprocessor with block diagram.
   b) Draw the functional architecture of a 8-bit microprocessor along with diagram and briefly describe about the interrupts available in it.
2. a) Draw and explain a well labelled timing diagram of the instruction MVI A, 32H and calculate the total execution time if clock frequency is 2 MHz.
   b) Write 8085 assembly programming language to find the smallest number among ten 8-bit data's stored in memory location 5000H to 5009H. Also store that value in 9100H and display on output device 51H.
3. a) Define T-state, Machine cycle and Instruction cycle. What are the different addressing models address in 8085 microprocessor?
   b) What are the three basic functions microprocessor should satisfy to interface any memory chip? Explain how 4096*8 PROM is interfaced with 8085 microprocessor.
4. a) Draw and explain the control word for 8255A PPI. Also, explain the different operating modes of 8255 PPI.
   b) Write an assembly language program in 8086 to display the string, "Computer Engineering" at console.
5. a) What do you understand by macros and procedures? State and explain the different types of assembler directives in 8086
   b) What are the different dedicated interrupt pointers in 8086 microprocessor? Draw and explain the block diagram of 8086 IVT table along with related memory address.
6. a) How DTE and DCE is interfaced with minimum lines using RS232 cable? Explain how two computers can be connected without modem.
   b) What are the different methods of parallel data communication? Explain in detail.
7. Write short notes on: (Any two) 2×5
   a) 8251- USART
   b) Maximum mode and minimum mode of 8086
   c) Microprocessor & Microcontroller

notes.arpanadhikari7.com.np
Available on notes.arpanadhikari7.com.np